

KYC-IDP

Multi-Agent Document Intelligence for Indian KYC

Tamper forensics · Regulatory compliance · Cross-document intelligence
Calibrated decisioning · Hash-chained audit

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1 Overview

KYC-IDP is a document intelligence platform for Indian identity documents (PAN and Aadhaar). It goes beyond field extraction to answer the questions that actually gate a KYC decision: is the document forged, does the packet satisfy regulation, do the documents agree with each other, and can the case be safely auto-approved. Every stage emits hash-chained audit events.

The system is built as a LangGraph state machine with dual-path extraction — a YOLOv8 + PaddleOCR fast path with a vision-language-model fallback — and is governed by an evaluation harness that gates CI on ratcheting quality thresholds.

1.1 Problem framing

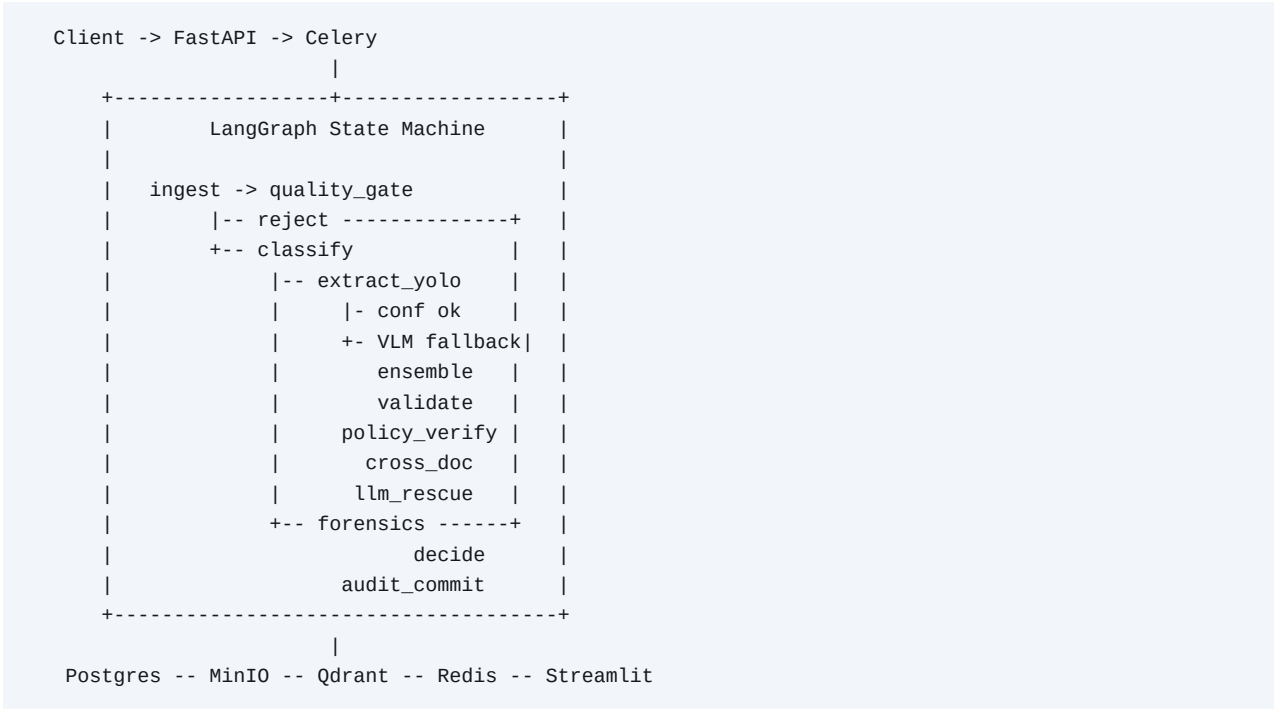
Question	Mechanism
Is this document forged?	Five-signal forensics suite, noisy-OR fusion
Does it satisfy RBI regulation?	Hybrid RAG with per-requirement citations
Do the documents agree?	Jaro-Winkler / Soundex entity resolution
Can we auto-approve safely?	Confidence calibration + hard overrides
Can we prove what happened?	SHA-256 hash-chained audit ledger
Are the metrics real?	Synthetic forge + held-out eval + CI gates

1.2 At a glance

Graph nodes	Unit tests	ADRs	Attack classes	Doc schemas
13	105	29	6	12+

2 Architecture

Processing is orchestrated as an explicit state machine rather than a linear call chain. Each stage is an independently testable node, routing is data-dependent, and forensics executes as a parallel branch so it never blocks extraction.



2.1 Dual-path extraction

Documents route to a fast detector-plus-OCR path when detection confidence is adequate, and to a vision-language model otherwise. An ensemble node selects between the two results. The consequence is graceful degradation: document types with no trained detector still extract, just more slowly.

Path	Pipeline	Typical latency
Fast	YOLOv8n detection → PaddleOCR on ROIs → normalize	~200 ms
Fallback	Vision-LLM (minicpm-v) structured extraction	~2–4 s

3 Measured Results

Forensics figures are produced by `make eval` and scored on a held-out seed pair (tuning=(42, 123), holdout=(777, 888), 10/type) never used for threshold tuning. They are reproducible from a clean checkout.

3.1 Forensics — held-out set

Metric	Result
Genuine false-positive rate	0 / 30 (0.0%)
Overall tamper recall	63.9%
Decision-layer leakage	0
Genuine auto-clear rate	100.0%

A KYC system must never flag a genuine customer. Zero false positives is a hard invariant, CI-enforced. Leakage counts forensically-flagged documents that nonetheless auto-cleared — a decision-layer bug class, gated at zero.

3.2 Recall by attack class

Attack	Recall	Detection basis
exif_edit	100.0%	EXIF software tags, date anomalies
text_splice	100.0% *	JPEG quantization-table quality
regenerate	100.0% *	JPEG quantization-table quality
screen_recapture	40.0%	FFT Moiré analysis
copy_move	43.3%	ORB keypoint matching (alignment-free)
font_swap	0.0%	Not yet detectable

* These two figures are tautological — read them carefully

The 100% recall on **text_splice** and **regenerate** reflects the detector firing on “*saved at quality < 88*”, which is an artifact of how the forge writes these attacks — not evidence of tampering. An adversary who re-saves at Q=92 evades detection entirely. The gates are set at 0.90 to catch regressions, **not** as a claim of adversarial robustness. Full analysis in ADR-028.

3.3 Detector latency

Detector	p50 (ms)	p95 (ms)
metadata	0.1	0.2
font	3.6	5.2
ELA	6.5	8.6
screen recapture	16.0	18.6
copy-move (ORB)	88.2	111.3

3.4 Field detection

Model	Source	mAP@50	mAP@50-95
Aadhaar	Pre-trained (HuggingFace)	0.963 †	0.748 †
PAN	Trained here (Roboflow data)	0.919 ‡	0.643 ‡

† Reported by the upstream model author; not independently re-measured.

3.5 PAN detector — per-class breakdown

Class	Precision	Recall	mAP@50
name	0.992	1.000	0.995
fathurname	0.933	1.000	0.995
dob	1.000	0.806	0.931
pan (the PAN number)	0.901	0.600	0.757

‡ PAN results are provisional — the aggregate hides the important finding

The PAN detector was trained on **71 images and validated on 6** (23 instances). An aggregate mAP over six images carries very wide error bars. More importantly, the headline 0.919 conceals that **pan** — the PAN number itself, the document's primary key — is the *weakest* class, missing 40% of instances. Training converged (box loss 2.21 → 0.93), so the ceiling is data volume, not schedule. A credible detector needs ≥500 annotated images merged across the available PAN datasets.

4 Scope and Validity of Results

Stated explicitly so the figures above are not over-read:

- All forensics numbers are measured on **synthetic** documents produced by the built-in Identity Forge — not production traffic.
- The 0% genuine false-positive rate holds against forge output saved at Q=92. Real submissions arrive via phone cameras, messaging apps and scanners at varied quality; real-world FPR is **unmeasured**.
- Decision-layer figures isolate the forensic gate — extraction, policy and cross-document scores are held at 1.0.
- PAN detection metrics rest on a six-image validation split and should be treated as directional only.
- End-to-end extraction F1 and full-graph p95 latency are **not yet benchmarked**.

The evaluation harness exists precisely to keep these claims falsifiable. It is what surfaced the tautology documented in §3.2 — the system reporting its own weakness rather than concealing it.

5 Component Reference

Module	Responsibility
services/graph/	LangGraph state machine — 13 nodes, conditional routing
services/forensics/	ELA, ORB copy-move, font, JPEG/EXIF metadata, FFT screen-recapture, noisy-OR scorer
services/rag/	Policy indexer, hybrid retriever (dense + BM25 + RRF), cross-encoder reranker, citation verifier
services/cross_doc/	Entity resolution, contradiction detection, Indian address normalization
services/decisioning/	Confidence calibrator, auto-clear engine with hard overrides
services/audit/	SHA-256 hash-chained ledger, state replay
services/active_learning/	Ground-truth store, retrain triggers, model registry, regression checker
services/extraction/	VLM extractor, ensemble scoring, normalizers, LLM cleaner

Module	Responsibility
tools/forge/	Identity Forge (Verhoeff-valid synthetic IDs) + Tamper Forge (6 attack classes)
tools/eval/	Evaluation harness with ratcheting CI gates

5.1 Technology

Layer	Technology	Layer	Technology
Orchestration	LangGraph	Vector store	Qdrant + bge-small-en-v1.5
API	FastAPI + Uvicorn	Reranking	ms-marco-MiniLM-L-6-v2
Queue	Celery + Redis	Detection	YOLOv8n (Ultralytics)
Vision LLM	minicpm-v (Ollama)	OCR	PaddleOCR
Text LLM	Qwen 3 8B (Ollama)	Persistence	Postgres + MinIO

6 Engineering Practices

6.1 Decision records

Twenty-nine ADRs capture every non-obvious decision, including the ones that failed. ADR-019 documents a rotation classifier that scored 0/4 on real cards and was disabled behind a config flag rather than shipped. ADR-028 documents the tautological detector in §3.2. Recording negative results is deliberate: it prevents the same ground being re-explored and keeps reported metrics anchored.

6.2 Ratcheting quality gates

config/eval_thresholds.yaml encodes the current *measured* floor rather than aspirations. Any change degrading a certified metric turns the build red. Floors are raised only after an eval run proves the improvement is real.

6.3 Held-out evaluation

Tuning and holdout seed pairs are separate and CI-enforced, so thresholds cannot be tuned into looking good on the data that produced them.

6.4 Independent audit per phase

Each phase is reviewed by a separate pass before the next begins, with findings classified Critical / Significant / Minor and remediated in place. The W5 audit surfaced six significant findings, all fixed prior to merge.

6.5 Failure modes made visible

Silent degradation is treated as a defect class in its own right. During W5 the PAN detector emitted class names absent from the configured field map; because unmapped labels are dropped by design, it produced zero fields and the pipeline quietly fell back to VLM-only — no exception, no error metric. A regression test now asserts that every class an installed model emits has a mapping, and was verified to fail against the broken configuration before being committed.

7 Known Gaps and Roadmap

Gap	Required work
PAN detector under-trained (71/6 split, pan recall 0%)	More available PAN datasets to ≥ 500 annotated images; re-train and re-validate
font_swap undetected (0% recall)	OCR-context font forensics — stroke-width and glyph-metric consistency
text_splice / regenerate detection evadable	Frequency-domain work: double-JPEG ghosts, DCT histogram analysis, learned localizer
No production-traffic calibration	Calibrate quality thresholds against real scan-quality distributions
Extraction F1 / end-to-end p95 unbenchmarked Extend the eval harness across the full graph	

7.1 Reproducing these results

```

docker compose up                # infrastructure
alembic upgrade head             # schema

.venv/bin/python -m pytest tests/unit -q    # 105 unit tests
make eval                            # regenerate all §3 figures

# field detectors
.venv/bin/python -m tools.train.download_pretrained --type aadhaar
read -s ROBOFLOW_API_KEY && export ROBOFLOW_API_KEY
.venv/bin/python -m tools.train.download_pretrained --type pan

```

Without detector weights present the pipeline runs VLM-only — functional, slower. Every figure in §3 regenerates from a clean checkout.